

Data-Free Neural Mesh Inpainting: Preserving Topology and Geometry

Haoliang Zhang , Jiali Zhang, Canran Xiao, Samuel Cheng, and Christian El Amm

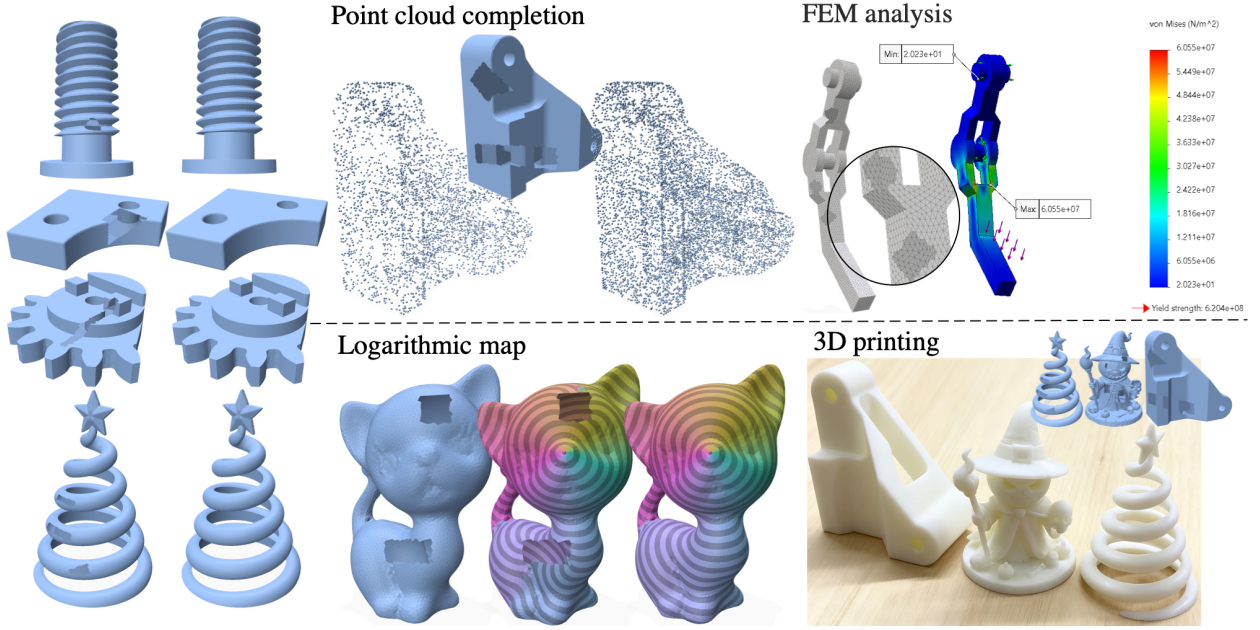


Fig. 1: Our data-free neural mesh inpainting method restores missing regions directly on the input mesh while preserving geometric and topological consistency. The repaired meshes can be seamlessly applied to downstream tasks such as point cloud completion, FEM simulation, logarithmic map computation, and 3D printing.

Abstract— Mesh inpainting remains a critical challenge in geometry processing, particularly for real-world scans that often contain holes and missing regions. This paper introduces a data-free neural mesh inpainting framework that operates directly on incomplete input meshes, preserving geometric and topological consistency. Our approach involves stitching corresponding regions from the zero-level isosurface and refining vertex positions using a self-supervised graph convolutional network. Extensive experiments demonstrate that our method outperforms traditional dataset-independent approaches and exhibits greater robustness than existing deep learning models. The repaired meshes maintain sufficient integrity for downstream applications such as physical simulation and 3D printing. Code is available at <https://github.com/ZHLRJ/Mesh-Inpainting>.

Index Terms—Intrinsic landmarks, Shape correspondence, Non-rigid shape matching

1 INTRODUCTION

High-quality and geometrically consistent meshes are indispensable for applications such as 3D printing, physical simulation, and computer graphics animation. However, meshes reconstructed from real-world data are often incomplete, exhibiting holes and missing regions caused by occlusion, reflective surfaces, and limited scanning coverage. Therefore, a reliable and effective repair algorithm is essential to bridge the gap between real-world reconstruction and downstream applications.

Mesh inpainting is an ill-posed problem—the missing geometry cannot be determined from the available data. For the missing regions, neither the vertex positions nor their connectivity information is available. Traditional methods [3, 14, 25, 40, 45, 67] rely primarily on geometric priors such as smoothness and self-similarity. While these

methods can effectively repair small or regularly shaped holes and preserve overall topology, they struggle to generalize to large or complex missing regions and often demand careful parameter tuning.

In contrast to purely geometry-based methods, recent works incorporate data-driven priors or apply supervised deep learning to improve completion quality. Some approaches developed for voxel grids [62] and point clouds [16, 30, 54, 58, 60, 64] ignore the intrinsic mesh connectivity. The primary challenge for point-cloud-based mesh repair lies in reliably reconstructing triangle connectivity. Even the latest generative models [10, 39, 49, 51, 66] typically produce meshes with only a few thousand faces, which limits their geometric fidelity. This constraint continues to hinder the advancement of mesh-based inpainting methods, even though large-scale 3D datasets are now available [28, 69]. More recent methods [24, 48, 55, 63] can produce high-quality meshes but rely on manually provided point clouds to guide the reconstruction of missing regions. Some self-prior-based methods do not require ground-truth inpainted geometries but still depend on large-scale shape datasets. These approaches [11, 12, 36] often alter the mesh format and change the topological properties of the input shape. These methods also perform poorly when encountering shapes that are

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not represented in the training datasets.

To address the above challenges, we present a Neural Mesh Inpainting (Fig. 1) that performs completion directly on the mesh domain. The geometric stage first projects the missing regions of the input mesh onto its zero-level isosurface to identify target areas for reconstruction. Based on this projection, a geodesic-driven triangulation algorithm enables cutting and stitching along arbitrary surface curves while maintaining valid topology. The process naturally supports parallel repair of multiple disconnected holes. Unlike previous approaches, our method eliminates the need for large-scale training datasets and offers a general, scalable solution applicable to arbitrary mesh geometries. In addition, the repaired meshes preserve the original structure, allowing seamless integration with existing neural architectures for fine-tuning.

Summary. In this paper, we present a data-free mesh inpainting method. Our approach is built around four key properties:

- Our method preserves the original mesh structure, modifying only the repaired regions while keeping the input geometry and topology completely intact.
- It scales efficiently to large and complex meshes, handling models with a high number of vertices and faces without compromising accuracy or stability.
- The approach operates entirely on a single input mesh without requiring any external training data.
- The repaired meshes can be directly applied to downstream tasks such as physical simulation and 3D printing, demonstrating its suitability for real-world use.

2 RELATED WORK

2.1 Traditional Geometry-Based Shape Repair

Traditional mesh inpainting methods can be broadly classified into surface-based and volume-based approaches. Surface-based methods reconstruct missing regions by generating new triangles guided by local geometric cues such as vertex density, triangle area, or dihedral angles [3, 14, 31, 40, 67]. They typically apply remeshing or smoothing to improve surface quality and perform well on small, simple holes, but often fail on large or topologically complex missing regions. To address these limitations, context-based methods [25, 45] enhance self-similarity by synthesizing features from similar patches, while Kraevoy and Sheffer [29] introduced a template-based approach that maps incomplete meshes to reference models. Although these extensions can better preserve geometric detail, they still depend on the availability of similar shapes, complete templates, and manual user interactions.

Volume-based methods first convert an input mesh into a discrete voxel representation and then perform hole filling within the volumetric domain. A common strategy is volumetric diffusion, which generates a complete voxelized volume from partial geometry [17, 21, 53]. Bischoff et al. [5] further proposed morphological dilation and erosion on an octree structure to efficiently fill holes in the voxel space. After completion, the surface mesh is reconstructed by extracting triangles from the isosurface. These approaches are advantageous for repairing complex surfaces and producing manifold meshes. However, they often cause unwanted deformation of nearby regions during diffusion or morphing, and their output resolution is constrained by the computational inefficiency of voxel-based representations.

2.2 Data-Driven Shape Inpainting

Data-driven shape inpainting methods used volumetric networks for shape recovery from partial scans [52, 62], while later works [16, 22] adopted multi-resolution prediction and coarse-to-fine refinement to improve reconstruction quality.

Point cloud completion has been widely investigated since point clouds are direct outputs of 3D scanners. Networks such as PCN [64] and SA-Net [58] reconstruct complete point clouds via encoder-decoder architectures, while GAN-based [9] and reinforcement learning approaches [44] further improve the realism of generated

shapes. PMP-Net [59] infers point trajectories to restore missing structures without explicit point generation. Recently, implicit neural representations have gained popularity for shape inpainting due to their memory efficiency and smooth surface reconstruction. Methods based on signed distance functions (SDF) [11, 38, 43] and vector-quantized implicit functions (VQDIF) [37, 63] have demonstrated strong shape completion capability.

Mittal et al. [36] trained a network to predict occluded regions in point clouds by simulating missing points from range scans, while Chu et al. [13] completed polygonal meshes through a truncated signed distance field (TSDF) representation. Although these methods eliminate the need for complete ground-truth shapes, they still depend on large-scale datasets such as ShapeNet [7], KITTI [4], and Thingi10k [69]. Moreover, converting meshes to point-based or volumetric forms often alters their topological properties, leading to inconsistencies in reconstructed geometry. When meshes are first converted into point clouds and later re-meshed, the generated surfaces are typically low-resolution and fail to preserve the original connectivity structure. A comparison summary can be found in Table 1. In contrast, our method operates directly on the original incomplete mesh without altering its structure. It preserves the topology and geometric continuity of the input while removing the dependency on external datasets, offering a more reliable and data-free solution for mesh inpainting.

2.3 Self-Prior Learning in Geometry Processing

Self-prior learning has emerged as a data-driven yet data-independent paradigm in geometry processing. The Deep Geometric Prior (DGP) [61] learns local surface representations from a single point cloud by training separate multilayer perceptrons (MLPs) for each region. Point2Mesh [24] extends this idea to the mesh domain through MeshCNN [23], which employs shared-weight convolutions and dynamic mesh pooling, and was later adapted for textured mesh reconstruction from colored point clouds [56]. For sparse or irregularly sampled point clouds, recent works have adopted unsupervised reconstruction techniques based on SDF [8, 34] and point consolidation strategies [35] to improve geometric consistency. The approach in [26] relies on MeshFix [2] for initial completion, and the final reconstruction quality is therefore largely determined by the preprocessing result rather than the learned model.

Despite their progress, existing self-prior methods remain restricted to point-based or volumetric representations and often reconstruct meshes from low-resolution surfaces, causing the loss of fine topology and structural connectivity. In contrast, we propose a data-free self-prior framework that operates directly on the input mesh, preserving its original geometry and topology throughout the inpainting process.

3 METHOD

In the following, we outline the main components of Neural MeshInpaint: a geometric stage that provides the tools for mesh clipping, stitching, and implanting, followed by a neural stage that fine-tunes the surface.

3.1 Geometric Preprocessing

Our domain is a triangle mesh $M = (V, E, F)$ with arbitrary topology. For an incomplete mesh, we first perform a preprocessing to obtain a watertight surface. The zero-level isosurface generated from the GSD [19] provides a contour (with contouring via marching cubes [33].) on which hole boundaries can be mapped, as shown in Fig. 2. However, this contour cannot be used directly, as it lacks geometric detail, is often non-manifold, and suffers from poor triangulation quality. Therefore, we refine it further and use it as an intermediate scaffold for subsequent processing.

To improve the quality of the initial isosurface mesh, we first apply a light QSLIM simplification [20] to remove spurious triangles. The remaining vertices are then processed with Poisson surface reconstruction [27] (octree depth of eight) to obtain a cleaner manifold surface. Finally, a QSLIM simplification followed by triangular remeshing [1, 47] ensures a vertex density comparable to the input mesh. The

Table 1: Method comparison. The Data-Free column indicates whether the method requires training datasets. Input Fidelity measures whether the original mesh structure is preserved. Scalability refers to the ability to handle large, high-resolution meshes. The Performance column reflects the overall reconstruction quality. Comments denote additional requirements such as real data or auxiliary tools.

	Data-Free	Output Type	Scalability	Input Fidelity	Performance	comment
PMP-Net [59]	✗	Point	✗	✗	Medium	
IF-Net [11]	✗	Point	✗	✗	Medium	
ShapeFormer [63]	✗	Point	✗	✗	Low	
SapceMesh [48]	✗	Mesh	✗	✗	High	Extra data need
SeMIGCN [26]	✓	Mesh	✓	✓	Medium	MeshFix init
MeshFix [2]	✓	Mesh	✓	✓	Low	
Ours	✓	Mesh	✓	✓	High	

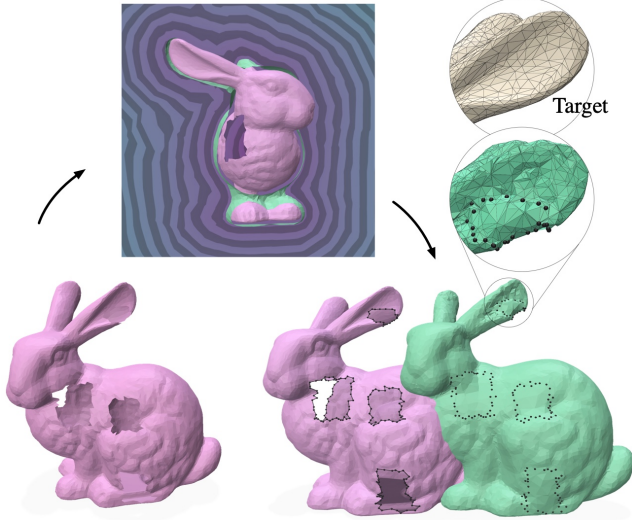
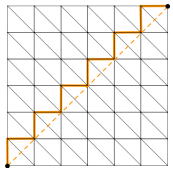


Fig. 2: A zero-level isosurface is extracted from the GSD (top), onto which the original hole boundary points are mapped. However, the low-quality mesh must be refined before mapping.

overall process, illustrated in Fig. 3, produces the target surface for boundary mapping and patch cutting.

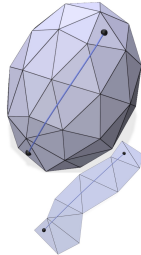
Generating cut lines: A simple way to generate cutting paths is to use traditional shortest path algorithms [18],



but such methods are constrained by mesh edges and often produce jagged paths (see inset). We instead use geodesic paths computed on the continuous surface, which yield smooth and geometry-consistent cuts.

A geodesic is the shortest path between two points constrained to lie on the surface; when the triangles along the path are unfolded, it appears as a straight line [41] (see inset). We compute

geodesics [46, 50] sequentially between consecutive boundary points and record the triangles they traverse. Projection points that fall inside a triangle are connected to its three vertices, splitting the face into three smaller ones. If two projection points lie in the same triangle, the subdivision is applied sequentially. For each triangle fully crossed by a geodesic, we select the crossing point on the intersected edge and connect it to the opposite vertex (Fig. 4). We leverage a half-edge [57] mesh representation to ensure edge-manifoldness and maintain consistent orientation. With these operations, the geodesic path is entirely represented by edges of the processed mesh. For holes that are topologically independent (i.e., sharing no common faces), our cutting algorithm can process them in parallel, further improving efficiency.



Each projected point x is parameterized using barycentric coordinates of triangle $ijk \in F$ with vertex positions v_i, v_j, v_k , as $x = t_i v_i + t_j v_j + t_k v_k$. We snap a projected point to the nearest edge if one barycentric weight is small, or to the corresponding vertex if two weights are small. This operation improves numerical stability and avoids low-quality triangles in the subsequent retriangulation step.

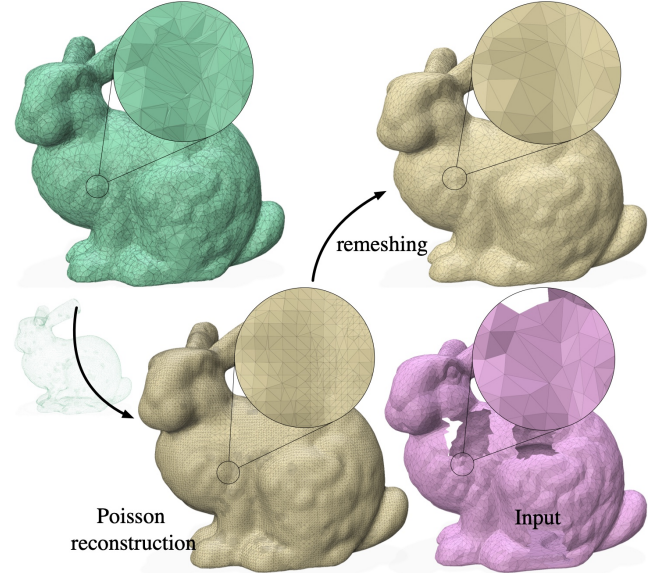


Fig. 3: Poisson surface reconstruction is applied to the initial isosurface to improve mesh quality, followed by remeshing to ensure a uniform vertex density comparable to the input mesh.

During the stitching process, we update the projected boundary vertices of each patch to their corresponding positions on the input mesh hole. For new vertices generated between two projected points, we insert them proportionally along the corresponding original edges, forming watertight connections between the patch and the input mesh.

3.2 Self-Supervision

Inspired by the mesh autoencoder architecture [6, 32, 65], we adopt it to learn feature representations of the input mesh using a graph convolution operator that avoids the intricate design of kernel functions. The learned features are then used to fine-tune the repaired regions for improved geometric consistency.

Notation. We denote a mesh as $\mathcal{M}$ with vertex and face sets $\mathcal{V}$ and $\mathcal{F}$. Vertex positions and face normals are represented as $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times 3}$ and $\mathbf{N} \in \mathbb{R}^{|\mathcal{F}| \times 3}$, respectively. The input hole mesh is denoted as $\mathcal{M}_{in}$, and the watertight mesh obtained after preprocessing (Sec. 3.1) is denoted as $\mathcal{M}_{init}$. The network outputs a completed mesh $\mathcal{M}_{cmp}$ with refined vertex positions at the target (hole-filled) regions. We define masks m_v^i and m_n^i to distinguish vertices and face normals in hole regions, where $m = 0$ for holes (real or fake) and $m = 1$ otherwise.

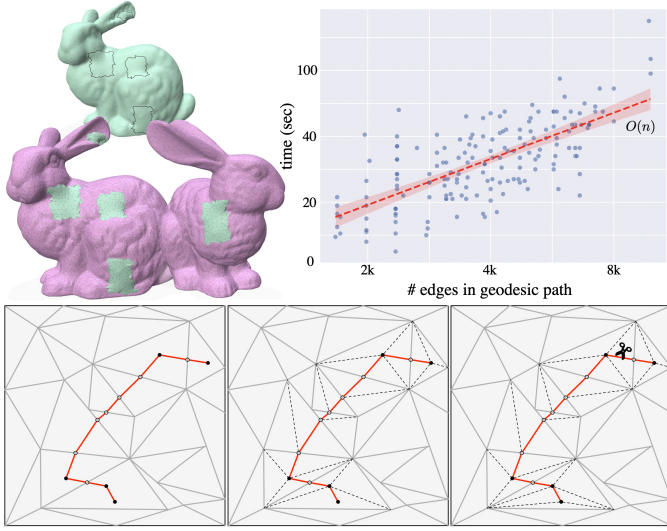


Fig. 4: Top: path stitching between the projection and input meshes. Bottom: retriangulating the faces intersected by the geodesic yields a valid cutting path (red) represented by mesh edges. The measured runtime (top-right) confirms a linear $O(n)$ relationship between path length and computation time. In this example, the bunny model contains 14K faces, with 400 edges along the geodesic path, and the entire cutting process takes 2.1 seconds.

Pooling/Unpooling. The model employs a mesh pooling operation based on QSLIM [20]. During pooling, vertex pairs are iteratively contracted to simplify the mesh while preserving geometric fidelity through quadric error metrics. The output features are obtained by multiplying the input features with a precomputed downsampling transformation matrix. The corresponding unpooling matrix is also stored for feature recovery during decoding [42].

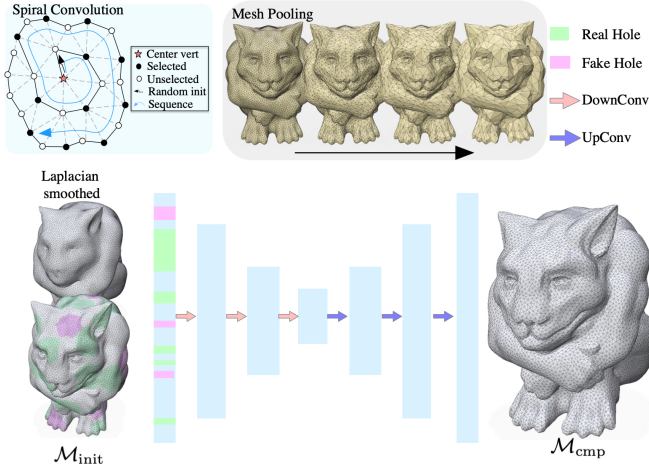


Fig. 5: Self-supervised training of our refinement model. Pooling and unpooling are implemented via QSLIM-based precomputed transformation matrices, consistent with prior mesh autoencoder designs [42]. The model predicts vertex displacements to reconstruct the inpainted surface from an oversmoothed mesh using both real (green) and fake (pink) holes.

Model Architecture. Fig. 5 illustrates self-supervised framework for mesh inpainting. Instead of directly predicting vertex positions, the Graph Convolutional Networks (GCN) estimates displacements from a Laplacian-smoothed mesh. The GCN takes per-vertex features $(\Delta x_{in}^i, \Delta y_{in}^i, \Delta z_{in}^i, m_v^i)$ as input, where $m_v^i \in \{0, 1\}$ indicates whether the vertex belongs to a real or fake hole. It predicts the 3D displacement

$(\Delta x_{out}^i, \Delta y_{out}^i, \Delta z_{out}^i)$ to recover the inpainted surface.

Loss Functions. The vertex position loss E_p is computed as the root mean squared error (RMSE) between the predicted and initial hole-filled mesh vertices, excluding real hole regions using the vertex mask m_v^i .

$$E_p = \left(\frac{1}{\sum_i m_v^i} \sum_{i=1}^{|V|} m_v^i \left\| \mathbf{x}_{cmp}^i - \mathbf{x}_{init}^i \right\|_2^2 \right)^{\frac{1}{2}}. \quad (1)$$

The facet normal loss E_n encourages local geometric fidelity by aligning the predicted and initial face normals, preserving sharp features. It is computed as a mean absolute difference over non-hole faces using the mask m_f^i , and evaluated only at the finest resolution to reduce computational cost:

$$E_n = \frac{1}{\sum_j m_f^j} \sum_{j=1}^{|F|} m_f^j \left\| \mathbf{n}_{cmp}^j - \mathbf{n}_{init}^j \right\|_1. \quad (2)$$

The smoothness term E_r regularizes facet normals to suppress noise and preserve sharp geometric features.

$$E_r = \frac{1}{|F|} \sum_{j=1}^{|F|} \left\| \mathbf{n}_{cmp}^j - S_{bnf}^{(t)}(\mathbf{n}_{cmp}^j) \right\|_1. \quad (3)$$

Here, $S_{bnf}^{(t)}$ denotes t iterations ($t = 5$ in this paper) of bilateral normal filtering (BNF) [68], which penalizes the deviation between predicted normals and their filtered counterparts.

Total loss. The overall training objective combines vertex, normal, and smoothness terms:

$$E = \sum \lambda_p E_p + \lambda_n E_n + \lambda_r E_r. \quad (4)$$

Each coefficient λ controls the relative contribution of the corresponding loss term. We use three down-pooling layers with $\lambda_p = 0.25$, and both λ_n and λ_r set to 0.5. Each simplification step reduces the number of vertices to 30% of the previous level.

4 RESULTS AND APPLICATIONS

We evaluate our method across a wide range of experiments. As shown in Figure 4, even for meshes with hundreds of thousands of triangles, our model processes geodesic paths spanning thousands of edges within tens of seconds, and only a few seconds in typical cases.

4.1 Accuracy and Performance

We first compare our method with three baseline approaches: the Advancing Front Method (ADVF) [67], MeshFix (MFX) [2], and Context-based Coherent Surface Completion (CCSC) [25]. All of these methods are data-free and operate directly on an input mesh. For intuitive visualization, we colorize the reconstructed meshes according to the distance between the output and the ground-truth surfaces.

Figure 6 presents representative qualitative results, while more comprehensive quantitative comparisons are provided in Tab. 2. For visualization purposes, the reconstruction errors are normalized and color-mapped onto the surface. As shown in this figure, traditional methods such as ADVF [67] and MFX [2] tend to fill holes with overly planar patches due to their reliance on local geometric heuristics. CCSC [25], while capable of producing smoother results, requires extensive optimization and takes over 500 seconds even for moderately sized meshes. In contrast, our method reconstructs the missing regions in just a few seconds, generating geometrically consistent surfaces that preserve the original curvature and fine-scale details.

4.2 Comparison with Supervised and Self-Prior Models

We further compare our approach with several recent data-driven models, including IF-Net [11], ShapeFormer [63], SeMIGCN [26], and SpaceMesh [48].

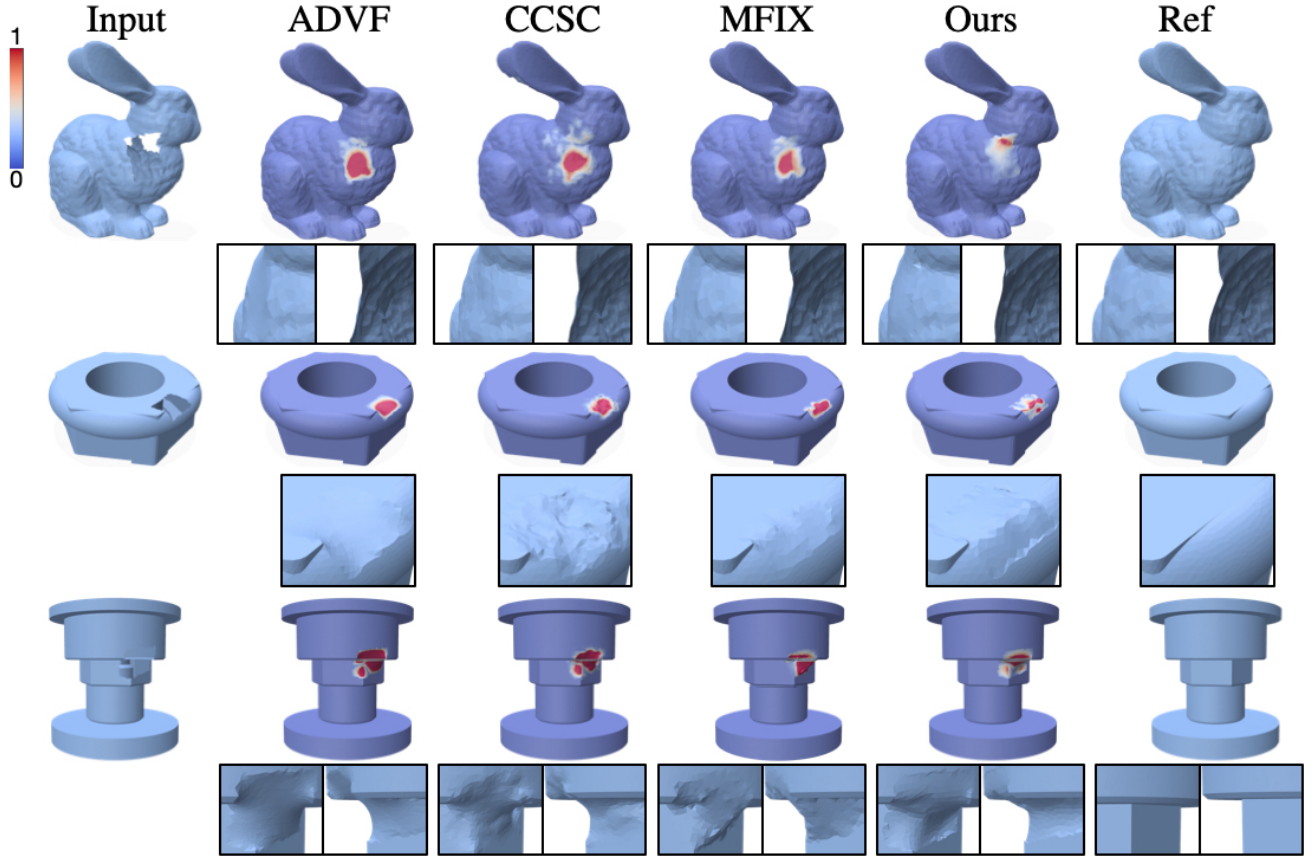


Fig. 6: Inpainting results on meshes with holes. The reconstructed meshes are colorized according to the distance to the ground-truth surfaces. Traditional geometry-based methods tend to produce noticeable bulges or inward dents near the repaired regions, while our method yields smoother and more accurate reconstructions.

Table 2: Quantitative comparison of inpainting performance using the average error distance in the repaired regions (in units of 10^{-3}).

	ADVF [67]	CCSC [25]	MFIX [2]	Ours
Bunny	6.01	5.31	5.55	4.33
Fandisk	5.69	4.94	2.78	0.54
Bimba	3.52	4.46	2.83	1.66
Bust	1.72	2.19	1.29	0.74
Igea	2.96	1.40	2.27	1.13
Draon	7.3	6.04	3.99	1.89
Nut	3.96	4.75	4.69	1.02

Figure 7 presents the results of these data-driven methods compared with our approach. For this comparison, we used the official pretrained weights and implementations provided by the respective authors of IF-Net and ShapeFormer.

For the transformer-based SpaceMesh [48], the model can only process meshes with a limited number of faces due to the inefficiency of its face tokenization algorithm. Moreover, it requires collaboration with other models or additional surface points around missing regions as conditions for mesh generation. SpaceMesh also alters the connectivity of the input mesh, resulting in structural inconsistencies.

More fundamentally, transformer-based architectures [10, 39, 49, 51] face intrinsic scalability limitations: constructing or processing meshes with tens of millions of faces is currently computationally infeasible due to the quadratic complexity of token attention and memory constraints.

All these data-driven methods fail to inpaint the missing geometries when the test meshes are not included in their training datasets. In contrast, our method is entirely independent of training data, operating

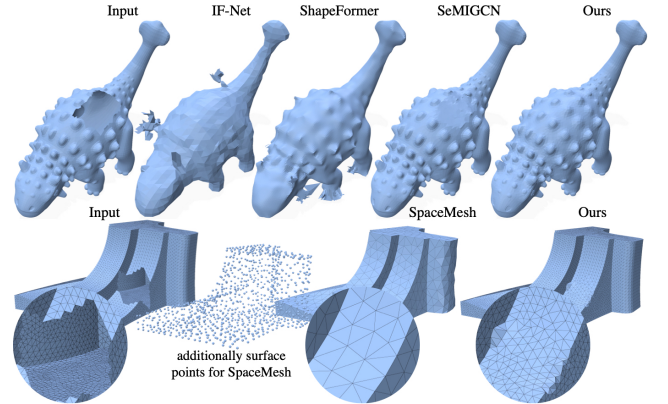


Fig. 7: Comparison with representative data-driven methods. While IF-Net and ShapeFormer produce coarse reconstructions and SeMIGCN failed to inpaint the shape geometries, SpaceMesh additionally requires surface point inputs. Our method achieves clean, detailed, and watertight repairs directly from the incomplete mesh.

robustly across arbitrary mesh inputs. SpaceMesh achieves nearly zero error because it requires the point cloud of the missing regions as input; thus, it only reconstructs connecting surfaces and is not directly comparable.

4.3 Applications

Our method enables several downstream applications that benefit from reliable and topology-preserving mesh repair. These applications high-

light the practicality and robustness of our approach for real-world geometric modeling and manufacturing workflows. We next illustrate several representative applications enabled by our repaired meshes.

Point Cloud. Our framework naturally serves as a bridge for point cloud completion. Given an incomplete point cloud, we first reconstruct a surface mesh, repair the missing regions using our mesh inpainting method, and then resample the completed mesh to obtain a dense, completed point cloud. Unlike data-driven completion methods, our approach requires neither large-scale training datasets nor costly model training, making it an efficient and data-free alternative for point cloud restoration.

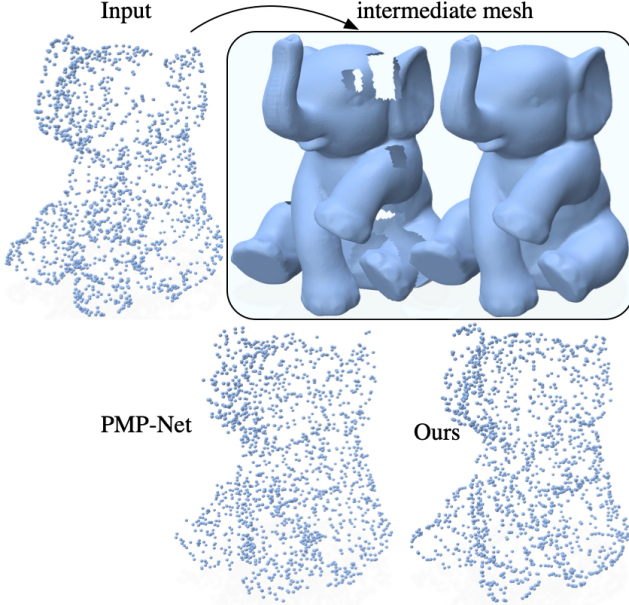


Fig. 8: Point cloud completion results. Our method reconstructs an intermediate mesh from the incomplete input, repairs it, and resamples a consistent point cloud.

Figure 8 shows a comparison with data-driven methods. PMP-Net [59] fails to recover missing regions for unseen shapes in its training data. Our data-free method, on the other hand, can inpaint arbitrary geometries while preserving natural shape continuity.

Simulation on Repaired Meshes. Our method preserves the original geometric and topological structure of the input mesh, making it directly compatible with downstream physical simulations. For example, in solving partial differential equations (PDEs) such as heat diffusion or deformation analysis on the surface, the repaired geometry can seamlessly inherit the original boundary conditions defined along the hole edges. Since our inpainting process maintains vertex connectivity and mesh consistency, simulation variables can be propagated continuously over the repaired regions without reinitializing the solver or remeshing the entire domain. This property enables efficient post-repair analysis and ensures that numerical stability and boundary fidelity are preserved.

To verify compatibility with downstream solvers, we perform surface PDE computations on the inpainted meshes. As shown in Fig. 9, both the logarithmic map and parallel transport [15] of tangent vectors can be seamlessly extended from the hole boundaries to the repaired regions without reinitialization. Since our method preserves the original connectivity and vertex indexing, the solver can reuse precomputed boundary conditions, enabling a continuous simulation workflow. On a mesh with 26K vertices and 52K faces, solving PDEs on the repaired surface requires only 5 ms, whereas recomputing from scratch takes 60 ms, highlighting the efficiency and physical consistency of our topology-preserving repair.

Additive Manufacturing Applications. In 3D printing, the repaired meshes generated by our method exhibit sufficient structural



Fig. 9: Surface PDE continuation on repaired meshes. Top: the input mesh with holes. Bottom: the repaired mesh. Logarithmic map and parallel transport are computed directly on the repaired surface by reusing boundary conditions along the hole edges.

integrity and watertightness to be fabricated directly. Because the surface continuity are preserved during inpainting, the repaired models can be exported as valid models without additional post-processing or remeshing.

Figure 10 demonstrates the use of the proposed mesh inpainting method for additive manufacturing. The blue models are the original meshes with missing regions, and the white models are the repaired and printed results. The repaired meshes maintain watertightness and structural stability, allowing direct fabrication without further processing.

5 CONCLUSION

We introduced a data-free neural mesh inpainting framework that reconstructs missing regions directly on the mesh domain while preserving the original topology and geometric consistency. Our approach integrates a topology-preserving geodesic cut-and-stitch algorithm for hole filling with a self-supervised graph convolutional network that refines vertex positions without external data. Through extensive evaluations, we demonstrated that our method achieves high reconstruction accuracy, strong robustness across diverse geometries, and seamless compatibility with downstream applications such as point cloud completion, physical simulation, and 3D printing. Future work includes extending the cut-and-stitch pipeline to dynamic meshes and integrating physical constraints for simulation-aware repair.

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Fig. 10: Repaired meshes from our method are watertight and structurally sound, allowing direct 3D printing without additional post-processing or remeshing.

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